MATHEMATICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

61. Let $A=\begin{bmatrix}\frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}}\\ \frac{-3}{\sqrt{10}} & \frac{1}{\sqrt{10}}\end{bmatrix}$ and $B=\begin{bmatrix}1 & -i\\ 0 & 1\end{bmatrix}$, where $i=\sqrt{-1}$.

If $M = A^TB\,A$, then the inverse of the matrix $AM^{2023}A^T$ is

(1) $\begin{bmatrix}1 & 0\\ -2023i & 1\end{bmatrix}$ (2) $\begin{bmatrix}1 & -2023i\\ 0 & 1\end{bmatrix}$

(3) $\begin{bmatrix}1 & 2023i\\ 0 & 1\end{bmatrix}$ (4) $\begin{bmatrix}1 & 0\\ 2023i & 1\end{bmatrix}$

Answer (2)

Sol. $\because \quad A=\begin{bmatrix}\frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}}\\ \frac{-3}{\sqrt{10}} & \frac{1}{\sqrt{10}}\end{bmatrix}$

$\therefore \quad A\cdot A^T = A^T\cdot A = I$

$\therefore \quad AM^{2023}A^T = B^{2023}$

$=\begin{bmatrix}1 & -2023i\\ 0 & 1\end{bmatrix}$

62. Let $\vec{a}=-\hat{i}-\hat{j}+\hat{k}$, $\vec{a}\cdot\vec{b}=1$ and $\vec{a}\times\vec{b}=\hat{i}-\hat{j}$. Then $\vec{a}-6\vec{b}$ is equal to

(1) $3\left(\hat{i}+\hat{j}-\hat{k}\right)$ (2) $3\left(\hat{i}-\hat{j}-\hat{k}\right)$

(3) $3\left(\hat{i}-\hat{j}+\hat{k}\right)$ (4) $3\left(\hat{i}+\hat{j}+\hat{k}\right)$

Answer (4)

Sol. $\vec{a}=-\hat{i}-\hat{j}+\hat{k}$ Let $\hat{b}=b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$

$\Rightarrow \quad -b_1-b_2+b_3=1 \quad \left(\vec{a}\cdot\vec{b}=1\right)$

&

$\vec{a}\times\vec{b}=-(b_2+b_3)\hat{i}+(b_1+b_3)\hat{j}-(b_1+b_2)\hat{k}=\hat{i}-\hat{j}$

$\Rightarrow \quad b_1+b_2=0,\ b_2+b_3=-1,\ b_1+b_3=-1$

also $\left|\hat{b}\right|=1$ (by Lami's Theorem)

$\Rightarrow \quad b_1=\frac{-2}{3},\ b_2=\frac{-2}{3}\ \&\ b_3=\frac{1}{3}$

$\Rightarrow \quad -6\hat{b}=4\hat{i}+4\hat{j}+2\hat{k}$

$\Rightarrow \quad \hat{a}-6\hat{b}=3\left(\hat{i}+\hat{j}+\hat{k}\right)$

63. Let $\Delta, \nabla \in \{\wedge, \vee\}$ be such that $(p\longrightarrow q)\,\Delta\,(p\,\nabla\,q)$ is a tautology. Then

(1) $\Delta=\vee,\ \nabla=\vee$ (2) $\Delta=\vee,\ \nabla=\wedge$

(3) $\Delta=\wedge,\ \nabla=\vee$ (4) $\Delta=\wedge,\ \nabla=\wedge$

Answer (2)

Sol. $(p\longrightarrow q)\,\Delta\,(p\,\nabla\,q)$

$\Rightarrow \quad (p'\vee q)\,\Delta\,(p\,\nabla\,q) \quad \ldots(i)$

If $\Delta=\vee,\ \nabla=\vee$

(i) becomes $(p'\vee q)\vee(p\vee q)\equiv T$

64. The equations of two sides of a variable triangle are $x = 0$ and $y = 3$, and its third side is a tangent to the parabola $y^2 = 6x$. The locus of its circumcentre is

(1) $4y^2 + 18y + 3x + 18 = 0$

(2) $4y^2 - 18y - 3x - 18 = 0$

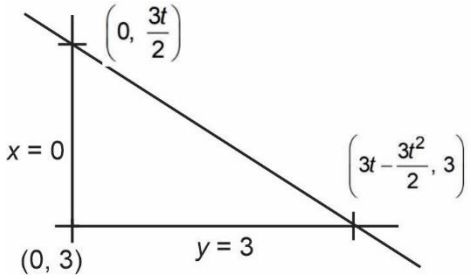
(3) $4y^2 - 18y + 3x + 18 = 0$

(4) $4y^2 - 18y - 3x + 18 = 0$

Answer (3)

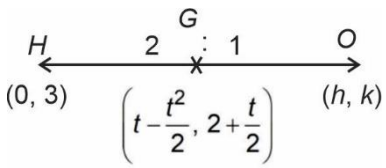
Sol. Third side of triangle

$ty=x+\frac{3}{2}t^2$



$\therefore \quad H\equiv(0,\ 3) \qquad G\equiv\left(t-\frac{t^2}{2},\ 2+\frac{t}{2},\right)$

Let $O(h, k)$



$\Rightarrow \frac{2h}{3} = t - \frac{t^2}{2}$ & $\frac{2k+3}{3} = 2 + \frac{t}{2}$

$\Rightarrow 4h = 6t - 3t^2$ & $4k = 6 + 3t$

$\Rightarrow 4h = 2(4k-6) - 3\left(\frac{(4k-6)^2}{9}\right)$

$\Rightarrow 3h = 6k - 9 - (4k^2 + 9 - 12k)$

$\Rightarrow 4k^2 - 18k + 3h + 18 = 0$

$\Rightarrow 4y^2 - 18y + 3x + 18 = 0$

65. The shortest distance between the lines $x + 1 = 2y = -12z$ and $x = y + 2 = 6z - 6$ is

(1) 2 (2) $\frac{3}{2}$

(3) 3 (4) $\frac{5}{2}$

Answer (1)

Sol. $L_1 : \frac{x+1}{1} = \frac{y}{\frac{1}{2}} = \frac{z}{\frac{-1}{12}}$

$L_2 : \frac{x}{1} = \frac{y+2}{1} = \frac{z-1}{\frac{1}{6}}$

$\text{S.D} = \left|\frac{\left(-\hat{i}+2\hat{j}-\hat{k}\right)\cdot\left(2\hat{i}-3\hat{j}+6\hat{k}\right)}{7}\right|$

$= \left|\frac{-2-6-6}{7}\right| = 2$ units

66. If the function

$$f(x) = \begin{cases} (1+|\cos x|)\frac{\lambda}{|\cos x|} & , \quad 0 < x < \frac{\pi}{2} \\ \mu & , \quad x = \frac{\pi}{2} \\ \frac{\cot 6x}{e^{\cot 4x}} & , \quad \frac{\pi}{2} < x < \pi \end{cases}$$

Is continuous at $x = \frac{\pi}{2}$, then $9\lambda + 6\log_e \mu + \mu^6 - e^{6\lambda}$ is equal to

(1) 11 (2) 8

(3) 10 (4) $2e^4 + 8$

Answer (3)

Sol. $\lim\limits_{x\to\frac{\pi}{2}^-} \left(1+|\cos x|\right)^{\frac{\lambda}{|\cos x|}} = e^{\lambda}$

$\lim\limits_{x\to\frac{\pi}{2}^+} e^{\frac{\cot 6x}{\cot 4x}} = \lim\limits_{x\to\frac{\pi}{2}^+} \frac{\tan 4x}{\tan 6x}$

$= e^{\frac{2}{3}}$

$\lambda = \frac{2}{3}, \ \mu = e^{\frac{2}{3}}$

$9\lambda + 6\ln\mu + \mu^6 - e^{6\lambda}$

$= 6 + 4 + e^4 - e^4 = 10$

67. If the four points, whose position vectors are $3\hat{i} - 4\hat{j} + 2\hat{k}$, $\hat{i} + 2\hat{j} - \hat{k}$, $-2\hat{i} - \hat{j} + 3\hat{k}$ and $5\hat{i} - 2\alpha\hat{j} + 4\hat{k}$ are coplanar, then α is equal to

(1) $\frac{107}{17}$ (2) $-\frac{107}{17}$

(3) $\frac{73}{17}$ (4) $-\frac{73}{17}$

Answer (3)

Sol. $\overrightarrow{AB} = -2\hat{i} + 6\hat{j} - 3\hat{k}$

$\overrightarrow{AC} = -5\hat{i} + 3\hat{j} + \hat{k}$

$\overrightarrow{AD} = 2\hat{i} + (4 - 2\alpha)\hat{j} + 2\hat{k}$

$\begin{vmatrix} -2 & 6 & -3 \\ -5 & 3 & 1 \\ 2 & 4-2\alpha & 2 \end{vmatrix} = 0$

$\Rightarrow 14b - 34\alpha = 0$

Or $\alpha = \frac{73}{17}$

68. Let T and C respectively be the transverse and conjugate axes of the hyperbola $16x^2 - y^2 + 64x + 4y + 44 = 0$. Then the area of the region above the parabola $x^2 = y + 4$, below the transverse axis T and on the right of the conjugate axis C is :

(1) $4\sqrt{6} + \frac{28}{3}$ (2) $4\sqrt{6} - \frac{44}{3}$

(3) $4\sqrt{6} + \frac{44}{3}$ (4) $4\sqrt{6} - \frac{28}{3}$

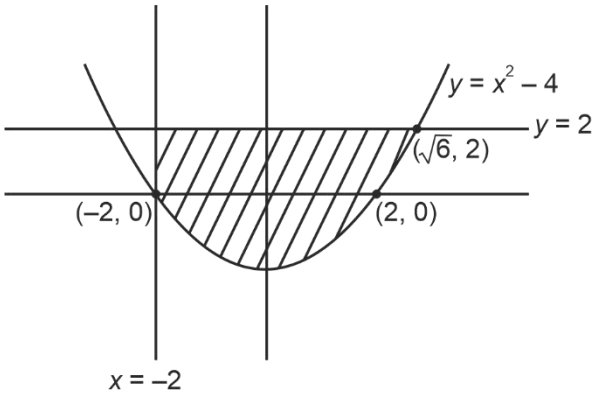
Answer (1)

Sol. $16(x+2)^2-(y-2)^2=16$

$$\frac{(x+2)^2}{1}-\frac{(y-2)^2}{16}=1$$

TA : $y=2$

CA : $x=-2$



$$A=\left|\int_{-2}^{\sqrt{6}}\left(2-\left(x^2-4\right)\right)dx\right|$$

$$=6x-\frac{x^3}{3}\Big|_{-2}^{\sqrt{6}}$$

$$=\left(6\sqrt{6}-\frac{6\sqrt{6}}{3}\right)-\left(-12+\frac{8}{3}\right)$$

$$=\frac{12\sqrt{6}}{3}+\frac{28}{3}$$

69. The number of numbers, strictly between 5000 and 10000 can be formed using the digits 1, 3, 5, 7, 9 without repetition, is

(1) 6 (2) 12

(3) 72 (4) 120

Answer (3)

Sol. $\underline{5}\ _\ _\ _ \Rightarrow {}^4C_3\cdot 3!=24$ ways

$\underline{7}\ _\ _\ _ \Rightarrow {}^4C_3\cdot 3!=24$ ways

$9\ _\ _\ _ \Rightarrow {}^4C_3\cdot 3!=24$ ways

Total ways = 72

70. Let $y=y(t)$ be a solution of the differential equation

$$\frac{dy}{dt}+\alpha y=\gamma e^{-\beta t}$$

where, $\alpha>0$, $\beta>0$ and $\gamma>0$. Then $\lim\limits_{x\to\infty} y(t)$

(1) Does not exist (2) Is 0

(3) Is –1 (4) Is 1

Answer (2)

Sol. $\frac{dy}{dt}+\alpha y=\gamma e^{-\beta t}$

I.F. $=e^{\int\alpha dt}=e^{\alpha t}$

$$\Rightarrow\quad y\cdot e^{\alpha t}=\gamma\int e^{(\alpha-\beta)t}dt=\gamma\frac{e^{(\alpha-\beta)t}}{(\alpha-\beta)}+C$$

$$\Rightarrow\quad y=\frac{\gamma}{(\alpha-\beta)}e^{-\beta t}+C\,e^{-\alpha t}$$

$$\lim_{x\to\infty} y(t)=\lim_{x\to\infty}\left[\frac{\gamma}{(\alpha-\beta)}e^{-\beta t}+C\,e^{-\alpha t}\right]=0$$

71. $\sum\limits_{k=0}^{6} {}^{51-k}C_3$ is equal to

(1) ${}^{52}C_3-{}^{45}C_3$ (2) ${}^{52}C_4-{}^{45}C_4$

(3) ${}^{51}C_3-{}^{45}C_3$ (4) ${}^{51}C_4-{}^{45}C_4$

Answer (2)

Sol. $\sum\limits_{k=0}^{6} {}^{51-k}C_3={}^{51}C_3+{}^{50}C_3+{}^{49}C_3+{}^{48}C_3+{}^{47}C_3$

$$+{}^{46}C_3+\left({}^{45}C_3+{}^{45}C_4\right)-{}^{45}C_4$$

$$S={}^{51}C_3+{}^{50}C_3+{}^{49}C_3+{}^{48}C_3+{}^{47}C_3+\left({}^{46}C_3+{}^{46}C_4\right)-{}^{45}C_4$$

$$\Rightarrow\quad S={}^{52}C_4-{}^{45}C_4$$

72. Let A, B, C be 3 × 3 matrices such that A is symmetric and B and C are skew-symmetric.

Consider the statements

(S1) $A^{13}B^{26}-B^{26}A^{13}$ is symmetric

(S2) $A^{26}C^{13}-C^{13}A^{26}$ is symmetric

Then,

(1) Both S1 and S2 are true

(2) Only S2 is true

(3) Only S1 is true

(4) Both S1 and S2 are false

Answer (2)

Sol. $A^T=A$, $B^T=-B$, $C^T=-C$

$P=A^{13}B^{26}-B^{26}A^{13}$

$$P^T=\left(A^{13}B^{26}-B^{26}A^{13}\right)^T=\left(A^{13}B^{26}\right)^T-\left(B^{26}A^B\right)^T$$

$$=\left(B^{26}\right)^T\left(A^{13}\right)^T-\left(A^{13}\right)^T\left(B^{26}\right)^T$$

$$=\left(B^T\right)^{26}\left(A^T\right)^{13}-\left(A^T\right)^{13}\left(A^T\right)^{26}$$

$$=B^{26}A^{13}-A^{13}B^{26}=-\left(A^{13}B^{26}-B^{26}A^{13}\right)=-P$$

P is skew-symmetric matrix $\Rightarrow S_1$ is false

$Q = A^{26}C^{13} - C^{13}A^{26} = Q^T = (A^{26}C^{13} - C^{13}A^{26})^T$

$Q = (A^{26}C^{13})^T - (C^{13}A^{26})^T = (C^{13})^T(A^{26})^T - (A^{26})^T(C^{13})^T$

$= (C^T)^{13}(A^T)^{26} - (A^T)^{26}(C^T)^{13} = -C^{13}A^{26} + A^{26}C^{13}$

$= A^{26}C^{13} + C^{13}A^{26}$

$\Rightarrow Q^T = Q \Rightarrow Q$ is symmetric matrix $\Rightarrow S_2$ is true.

73. Let $f(x) = 2x^n + \lambda, \lambda \in \mathbb{R}, n \in \mathbb{N}$ and $f(4) = 133$, $f(5) = 255$. Then the sum of all the positive integer divisors of $(f(3) - f(2))$ is

(1) 61 (2) 59

(3) 60 (4) 58

Answer (3)

Sol. $f(x) = 2x^n + \lambda, \lambda \in \mathbb{R}, n \in \mathbb{N}$

$f(4) = 2 \cdot 4^n + \lambda = 133,\ f(5) = 2 \cdot 5^n + \lambda = 255$

$f(5) - f(4) = 2 \cdot (5^n \cdot 4^n) = 122 \Rightarrow n = 3$

$\Rightarrow f(3) - f(2) = 2 \cdot (3^n \cdot 2^n) = 2 \cdot (3^3 - 2^3) = 2 \times 19$

Required sum = 1 + 2 + 19 + 38 = 60

74. The number of functions

$f : \{1, 2, 3, 4\} \to \{a \in : \mathbb{Z} \mid \alpha \mid \le 8\}$

satisfying $f(n) + \frac{1}{n} f(n+1) = 1, \ \forall \ n \in \{1, 2, 3\}$ is

(1) 3 (2) 2

(3) 4 (4) 1

Answer (2)

Sol. $\because \quad f : \{1, 2, 3, 4\} \to \{a \in \mathbb{Z} : |9| \le 8\}$

and $f(n) + \frac{1}{n} f(n+1) = 1$

$\Rightarrow \quad n\, f(n) + f(n+1) = n \quad \ldots(i)$

$\therefore \quad f(1) + f(2) = 1 \Rightarrow f(2) = 1 - f(1)$

But $f(1) \in [-8, 8]$

Hence, $f(2) \in [-8, 8] \Rightarrow f(1) \in [-7, 8] \ldots(A)$

and $2f(2) + f(3) = 2 \Rightarrow f(3) = 2f(1)$

$\therefore \quad 2f(1) \in [-8, 8] \Rightarrow f(1) \in [-4, 4] \ldots(B)$

and $3f(3) + f(4) = 3 \Rightarrow f(4) = 3 - 6f(1)$

$\therefore \quad f(1) \in \left[-\frac{5}{6}, \frac{11}{6}\right] \quad \ldots(C)$

From (A), (B) and (C) : f(1) = 0 or 1

$\therefore$ Only two functions are possible.

75. Let the function $f(x) = 2x^3 + (2p - 7)x^2 + 3(2p - 9)x - 6$ have a maxima for some value of $x < 0$ and a minima for some value of $x > 0$. Then, the set of all values of p is

(1) $\left(0, \frac{9}{2}\right)$ (2) $\left(\frac{9}{2}, \infty\right)$

(3) $\left(-\frac{9}{2}, \frac{9}{2}\right)$ (4) $\left(-\infty, \frac{9}{2}\right)$

Answer (4)

Sol. $f'(x) = 6x^2 + 2x(2p - 7) + 3(2p - 9)$

$x_1 < 0,\ x_2 > 0$

$\Rightarrow \quad f'(0) < 0$

$\Rightarrow \quad p < \frac{9}{2}$

76. The integral $16\int_1^2 \frac{dx}{x^3(x^2+2)}$ is equal to

(1) $\frac{11}{12} + \log_e 4$ (2) $\frac{11}{6} - \log_e 4$

(3) $\frac{11}{12} - \log_e 4$ (4) $\frac{11}{6} + \log_e 4$

Answer (2)

Sol. $I = \int \frac{dx}{x^3(x^2+2)^2}$

$= \frac{1}{4}\int \frac{x}{x^2+2}\,dx + \frac{1}{4}\int \frac{x}{(x^2+2)^2} - \frac{1}{4}\int \frac{dx}{x} + \frac{1}{4}\int \frac{dx}{x^3}$

$= \frac{1}{8}\ln(x^2+2) - \frac{\ln x}{4} - \frac{1}{8(x^2+2)} - \frac{1}{8x^3}$

Now, $16\int_1^2 \frac{dx}{x^3(x^2+2)^2} = 2\ln 6 - 2\ln 3 - 4\ln 2 + \frac{11}{6}$

$= \frac{11}{6} - \ln 4$

77. Let $f : R \to R$ be a function defined by $f(x) = \log_{\sqrt{m}}\{\sqrt{2}(\sin x - \cos x) + m - 2\}$, for some m, such that the range of f is [0, 2]. Then the value of m is _____

(1) 4 (2) 2

(3) 3 (4) 5

Answer (4)

Sol. We know that $\sin x - \cos x \in [-\sqrt{2}, \sqrt{2}]$

$$\log_{\sqrt{M}}\left(\sqrt{2}(\sin x - cox) + M - 2\right)$$

$$\in \left[\log_{\sqrt{M}}(M-4), \log_{\sqrt{M}} M\right]$$

$\Rightarrow \log_{\sqrt{M}}(M-4) = 0 \Rightarrow M = 5$

78. Let N be the sum of the numbers appeared when two fair dice are rolled and let the probability that $N-2, \sqrt{3N}, N+2$ are in geometric progression be $\frac{k}{48}$. Then the value of k is

(1) 16 (2) 8

(3) 4 (4) 2

Answer (3)

Sol. $n-2, \sqrt{3n}, n+2 \rightarrow$ G.P.

$3n = n^2 - 4$

$\Rightarrow n^2 - 3n - 4 = 0$

$\Rightarrow n = 4, -1$ (rejected)

$$P(S=4) = \frac{3}{36} = \frac{1}{12} = \frac{4}{48}$$

$\therefore k = 4$

Option (3) is correct.

79. Let z be a complex number such that $\left|\frac{z-2i}{z+i}\right| = 2, z \neq -i$. Then z lies on the circle of radius 2 and centre

(1) (0, 2) (2) (2, 0)

(3) (0, 0) (4) (0, –2)

Answer (4)

Sol. $\left|\frac{z-2i}{z+i}\right| = 2$

$\Rightarrow (z-2i)(\bar{z}+2i) = 4(z+i)(\bar{z}-i)$

$\Rightarrow 2\bar{z} + 2iz - 2i\bar{z} + 4 = 4\left(z\bar{z} - zi + \bar{z}i + 1\right)$

$\Rightarrow 3z\bar{z} - 6iz + 6i\bar{z} = 0$

$\Rightarrow 2\bar{z} - 2iz + 2i\bar{z} = 0$

$\therefore$ Centre $(-2i)$ or $(0, -2)$

Option (4) is correct.

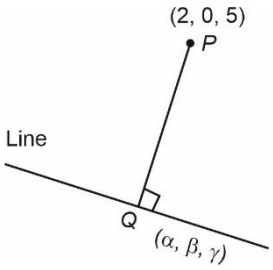
80. The foot of perpendicular of the point (2, 0, 5) on the line $\frac{x+1}{2} = \frac{y-1}{5} = \frac{z+1}{-1}$ is (α, β, γ). Then, which of the following is NOT correct?

(1) $\frac{\alpha\beta}{\gamma} = \frac{4}{15}$ (2) $\frac{\beta}{\gamma} = -5$

(3) $\frac{\gamma}{\alpha} = \frac{5}{8}$ (4) $\frac{\alpha}{\beta} = -8$

Answer (2)

Sol. $L: \frac{x+1}{2} = \frac{y-1}{5} = \frac{z+1}{-1} = t$



then,

$\alpha = 2t - 1$

$\beta = 5t + 1$

$\gamma = -t - 1$

for foot of $\perp^r$ $2(2t-3) + 5(5t+1) - (-t-6) = 0$

$\Rightarrow 30t + 5 = 0$

$\therefore t = -\frac{1}{6}$

$\therefore \alpha = \frac{-1}{3} - 1 = \frac{-4}{3}, \beta = \frac{1}{6}, \gamma = \frac{-5}{6}$

So, $\frac{\beta}{\gamma} = \frac{\frac{1}{6}}{-\frac{5}{6}} = \frac{-1}{5}$

So, option (2) is correct.

SECTION - B

Numerical Value Type Questions: This section contains 10 questions. In Section B, attempt any five questions out of 10. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the second decimal place; e.g. 06.25, 07.00, –00.33, –00.30, 30.27, –27.30) using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.

81. Let $a \in \mathbb{R}$ and let α, β be the roots of the equation $x^2 + 60^{\frac{1}{4}}x + a = 0$. If $\alpha^4 + \beta^4 = -30$, then the product of all possible values of a is _____

Answer (45)

Sol. $x^2 + 60^{\frac{1}{4}}x + a = 0$

$\therefore \quad \alpha + \beta = -60^{\frac{1}{4}}, \alpha\beta = a$

Now $\alpha^4 + \beta^4 = -30$

$\Rightarrow \quad (\alpha^2 + \beta^2)^2 - 2a^2 = -30$

$\Rightarrow \quad \left[(\alpha+\beta)^2 - 2a\right]^2 - 2a^2 = -30$

$\Rightarrow \quad \left(60^{\frac{1}{2}} - 2a\right)^2 - 2a^2 = -30$

$\Rightarrow \quad 60 + 4a^2 - 4 \cdot 60^{\frac{1}{2}}a - 2a^2 + 30 = 0$

$\Rightarrow \quad 2a^2 - 8\sqrt{15}a + 90 = 0$

Product of value of $a = 45$

82. For the two positive number a, b, if a, b and $\frac{1}{18}$ are in a geometric progression, while $\frac{1}{a}$, 10 and $\frac{1}{b}$ are in arithmetic progression, then $16a + 12b$ is equal to _____.

Answer (03)

Sol. $\because \quad a, b, \frac{1}{18} \rightarrow$ G.P.

$\therefore \quad b^2 = \frac{a}{18} \qquad \ldots(1)$

And $\frac{1}{a}, 10, \frac{1}{b} \rightarrow$ A.P.

$\therefore \quad 20 = \frac{1}{a} + \frac{1}{b}$

$20ab = a + b$

By (1) $a = 18b^2$

$\therefore \quad 20 \times 18b^3 = 18b^2 + b$

$\because \quad a, b > 0$

$360b^2 - 18b - 1 = 0$

$\Rightarrow \quad 360b^2 - 30b + 12b - 1 = 0$

$\Rightarrow \quad 30b(12b - 1) + 1(12b - 1) = 0$

$b = \frac{1}{12}, b = \frac{-1}{30}x$

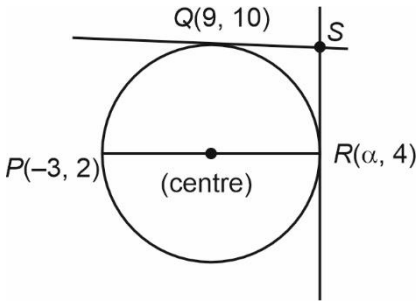
$\therefore \quad 12b = 1, a = 18 \times \frac{1}{144} = \frac{2}{16}$

$\therefore \quad 16a + 12b = 3$

83. Points $P(-3, 2)$, $Q(9, 10)$ and $R(\alpha, 4)$ lie on a circle C with PR as its diameter. The tangents to C at the points Q and R intersect at the point S. If S lies on the line $2x - ky = 1$, then K is equal to

Answer (03)

Sol.



Now, $\frac{10-2}{9+3} \times \frac{10-4}{9-\alpha} = -1$

$\Rightarrow \quad \frac{8}{12}.6 = \alpha - 9 \Rightarrow \boxed{\alpha = 13}$

$\therefore \quad 0 = (5, 3)$ So, $m_{OQ} = \frac{7}{4}$, $m_{OR} = \frac{1}{8}$

$\therefore \quad Q : y - 10 = \frac{-4}{7}(x - 9)$

$\Rightarrow \quad 4x + 7y = 106 \qquad \ldots(i)$

Tangent at R : $y - 4 = -8(x - 13)$

$8x + y = 108 \qquad \ldots(ii)$

By (i) and (ii) $S = \left(\frac{25}{2}, 8\right)$, satisfies with the line

$\therefore \quad \boxed{K = 3}$

84. If the shortest distance between the line joining the points (1, 2, 3) and (2, 3, 4), and the line $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-2}{0}$ is α, then $28\alpha^2$ is equal to _____.

Answer (18)

Sol. Points (1, 2, 3) and (2, 3, 4)

$L_1 : \frac{(x-1)}{1} = \frac{(y-2)}{1} = \frac{(2-3)}{1}$

$L_2 : \frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-2}{0}$

$\vec{b}_1 = \hat{i} + \hat{j} + \hat{k}$

$\vec{b}_2 = 2\hat{i} - \hat{j} + 0\hat{k}$

$\vec{a_1} - \vec{a_2} = 0\hat{i} - 3\hat{j} - \hat{k}$

$$d = \left|\frac{(\vec{a}_1 - \vec{a}_2)\cdot(n_1 \times n_2)}{|n_1 \times n_2|}\right|$$

$$= \left|\frac{6-3}{\sqrt{9+1+4}}\right| = \frac{3}{\sqrt{14}} = \alpha$$

$$28\alpha^2 = \frac{28\times 9}{14} = 18$$

85. Suppose Anil's mother wants to give 5 whole fruits to Anil from a basket of 7 red apples, 5 white apples and 8 oranges. If in the selected 5 fruits, at least 2 oranges, at least one red apple and at least one white apple must be given, then the number of ways, Anil's mother can offer 5 fruits to Anil is ____

Answer (6860)

Sol. Total 8 oranges, 5 white apple and 7 red apple.

5 fruits needs to be selected.

Case I : 3 orange + 1 red apple + 1white apple

$= {}^8C_3 \times {}^7C_1 \times {}^5C_1 = 1960$

Case II : 2 oranges + 2 red apples + 1 white apple.

$= {}^8C_2 \times {}^7C_2 \times {}^5C_1 = 2940$

Case III : 2 oranges + 1 red apples + 2 white apple.

$= {}^8C_2 \times {}^7C_1 \times {}^5C_2$

$= 1960$

Total = 1960 + 2940 + 1960

= 6860

86. 25% of the population are smokers. A smoker has 27 times more chances to develop lung cancer than a non-smoker. A person is diagnosed with lung cancer and the probability that this person is a smoker is $\frac{k}{10}$. Then the value of k is _____.

Answer (09)

Sol. Probability of a person being smoker = $\frac{1}{4}$

Probability of a person being non-smoker = $\frac{3}{4}$

$$P\left(\frac{\text{Person is smoker}}{\text{Person diagonsed with cancer}}\right) = \frac{\frac{1}{4}\cdot 27P}{\frac{1}{4}\cdot 27P + \frac{3P}{4}}$$

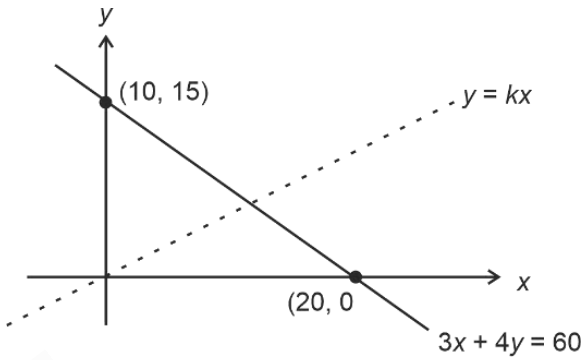
$$= \frac{9}{10} = \frac{k}{10}$$

$\Rightarrow \boxed{k = 9}$

87. A triangle is formed by X-axis and the line $3x + 4y = 60$. Then the number of points $P(a, b)$, which lie strictly inside the triangle, where a is integer and b is a multiple of a, is _____.

Answer (31)

Sol.



As b is multiple of a the required point lie on

Line $y = kx$ $(k \in z)$

$\therefore$ $3x + 4kx = 60$

$$x = \frac{60}{3+4k}$$

If $k = 1$	8 integral points
$k = 2$	5 integral points
$k = 3$	3 integral points
$k = 4$	3 integral points
$k = 5$	2 integral points
$k = 6$	2 integral points
$k = 7$	1 integral point
$k = 8$	1 integral point
⋮	⋮
$k = 14$	1 integral point

$\therefore$ Total 31 points

88. The remainder when $(2023)^{2023}$ is divided by 35 is _____.

Answer (07)

Sol. Let $N = 2023$

2023 is divisible by 7

$\therefore$ 2023^{2023} is divisible by 7

$\therefore$ Let $N = 7\alpha$

$$N = 2023^{2023} \equiv 3^{2023} \pmod 5$$

$$\equiv 3^3 \pmod 5 \equiv 2 \pmod 5$$

$\therefore$ $N = 5\beta + 2$

$\Rightarrow 7\alpha = 5\beta + 2$

$7\alpha = 5\beta + 7 - 5$

$7(\alpha - 1) = 5(\beta - 1)$

$\alpha - 1$ is divisible by 5

$\alpha = 5p + 1$

$N = 7\alpha = 7(5p + 1) = 35p + 7$

89. If m and n respectively are the numbers of positive and negative values of θ in the interval $[-\pi, \pi]$ that satisfy the equation $\cos 2\theta \cos\frac{\theta}{2} = \cos 3\theta \cos\frac{9\theta}{2}$, then mn is equal to _____.

Answer (25)

Sol. $2\cos 2\theta \cos\frac{\theta}{2} = 2\cos 3\theta \cos\frac{9\theta}{2}$

$\cos\frac{5\theta}{2} + \cos\frac{3\theta}{2} = \cos\frac{15\theta}{2} + \cos\frac{3\theta}{2}$

$\cos\frac{5\theta}{2} = \cos\frac{15\theta}{2}$

$\frac{15\theta}{2} = 2n\pi \pm \frac{5\theta}{2}$

$\frac{15\theta}{2} \pm \frac{5\theta}{2} = 2n\pi$

$10\theta = 2n\pi$ or $5\theta = 2n\pi$

$\theta = \frac{n\pi}{5}$ or $\theta = \frac{2n\pi}{5}$

$\Rightarrow \theta = \frac{n\pi}{5}$

$\theta = \pm\pi, \pm\frac{4\pi}{5}, \pm\frac{3\pi}{\pi}, \pm\frac{2\pi}{5}, \pm\frac{\pi}{5}$

$m = 5, \quad n = 5$

$mn = 25$

90. If $\int_{\frac{1}{3}}^{3} |\log_e x|\, dx = \frac{m}{n}\log_e\left(\frac{n^2}{e}\right)$, where m and n are coprime natural numbers, then $m^2 + n^2 - 5$ is equal to _____.

Answer (20)

Sol. $I = \int_{\frac{1}{3}}^{3} |\ln x|\, dx = -\int_{\frac{1}{3}}^{1} \ln x\, dx + \int_{1}^{3} \ln x\, dx$

$= -[x\ln x - x]_{\frac{1}{3}}^{1} + x\ln x - x]_{1}^{3}$

$= -\left[(0-1) - \left(\frac{1}{3}\ln 3 - \frac{1}{3}\right)\right] + \left[(3\ln 3 - 3) - (0-1)\right]$

$= \frac{2}{3} - \frac{1}{3}\ln 3 + 3\ln 3 - 2$

$= \frac{8}{3}\ln 3 - \frac{4}{3}$

$= \frac{4}{3}(2\ln 3 - \ln e)$

$= \frac{4}{3}\ln\left(\frac{3^2}{e}\right)$

$m = 4, m = 3$

$m^2 + n^2 - 5 = 20$

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